Installation instructions

Instructions

We will mainly use the R-packages rjags or R-NIMBLE for Bayesian modeling. If there is interest, the use of R2OpenBUGS can also be used. It would be good to install rjags and/or R-NIMBLE on your laptop. The instructions are below.

In addition, to summarize the results from a Bayesian analysis, some other R-packages are useful. These can be installed using the following code in R:

install.packages(c("readr","coda","runjags","MCMCvis","ggmcmc","basicMCMCplots"))

Installation R-package rjags

First install JAGS:

1) Go to website http://mcmc-jags.sourceforge.net (http://mcmc-jags.sourceforge.net/) for instructions. To download the

JAGS installation program (JAGS –x.x.x.exe), visit the webpage https://sourceforge.net/projects/mcmc-jags/files.

2) Run the JAGS installation program to install JAGS. When asked to Choose Install

Location, select the folder created earlier, e.g. C:\Bayesian\JAGS-4.2.0,

and follow the installation instructions.

Then, you can install the R-packages rjags (to let R talk to JAGS). There are

two ways to do this:

1) In the RStudio menu bar at the top of the screen, select Tools → Install Packages, then

type rjags under Packages

or

2) In the R Studio console window, type install.packages("rjags")

Installation R-NIMBLE

See instructions for windows and Mac users on:

https://r-nimble.org/html_manual/cha-installing-nimble.html (https://r-nimble.org/html_manual/cha-installing-nimble.html)

Note: You first need to install RTools On Windows you can install Rtools.exe from here (http://cran.r-project.org/bin/windows/Rtools).

Important: with R version 4.0 or newer, for which you need Rtools40, be sure to follow the instructions in the section "Putting Rtools on the PATH" to set the PATH after the installation is complete. For older versions of R, for which you would use Rtools35.exe, be sure to check the box that will modify your PATH during the installation process.

Installation R-package R2OpenBUGS (on Windows only)

First download and install OpenBugs on https://www.mrc-bsu.cam.ac.uk/software/bugs/openbugs/ (https://www.mrc-bsu.cam.ac.uk/software/bugs/openbugs/)

Install the latest version of this R package by entering the following in R:

install.packages("R2OpenBUGS")